



Food and Agriculture Organization
of the United Nations



Somalia Water and Land Information Management

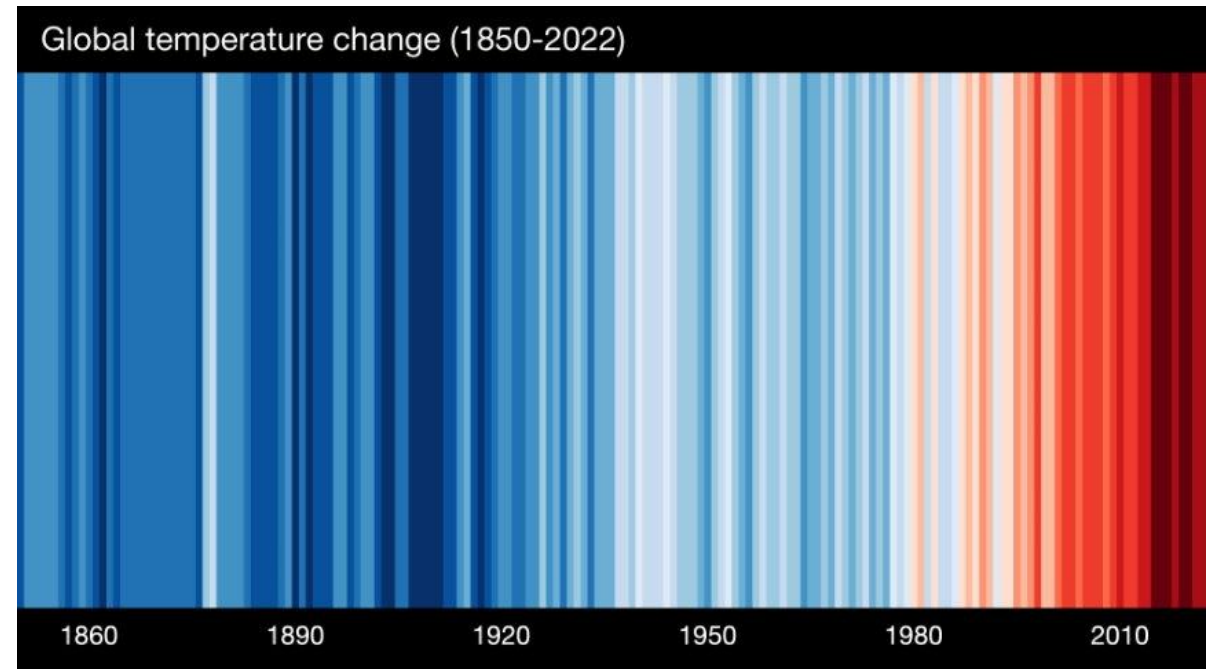
From Land and Water Geo-Analytics to impactful AA

Livia Peiser, Head of Geospatial Unit, Land and Water Division
(LTO for SWALIM project)

Agriculture and water in a changing climate

Climate crisis is a water crisis

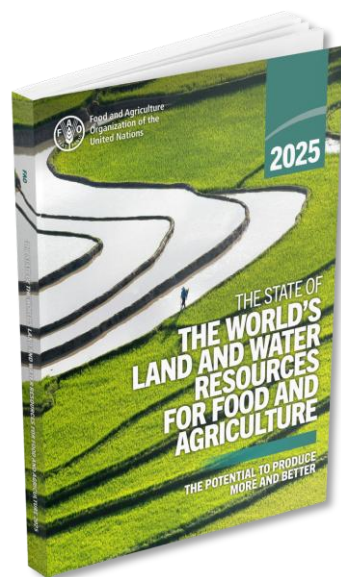
Increasing temperatures are
altering the hydrological cycle
and fragile countries face
recurring series of drought and
floods



Land degradation amplifies impacts on agriculture

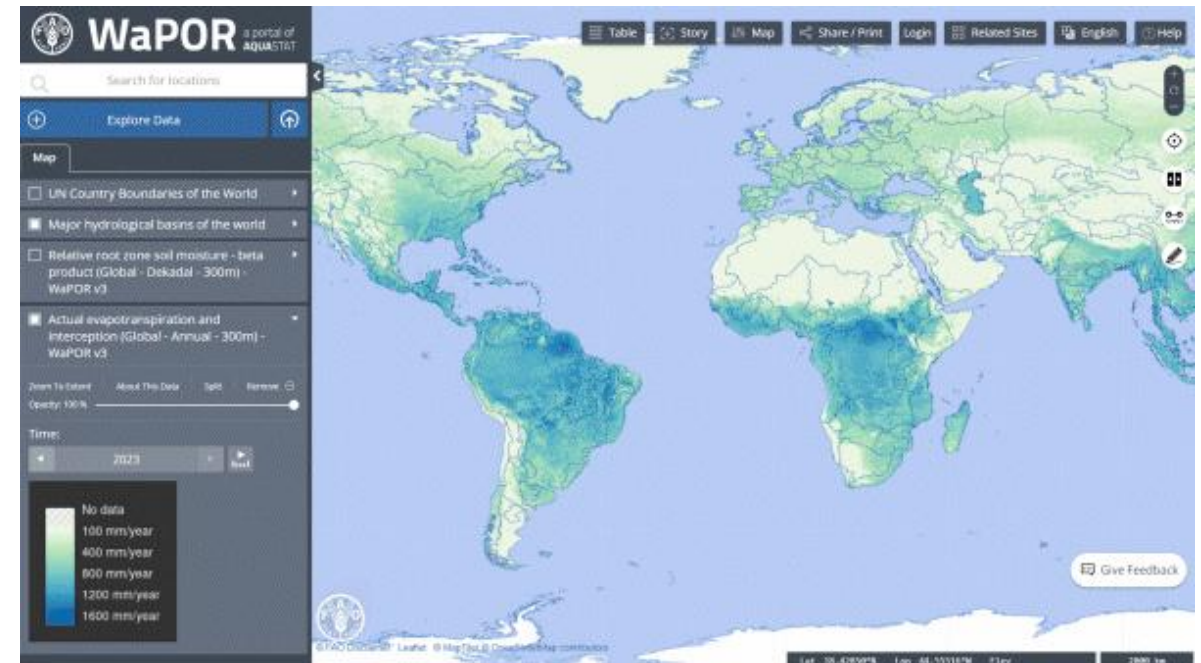
Drought is the costliest disaster for agriculture

It accounts for over one-third of all crop and livestock production losses caused by disasters in low- and middle-income countries



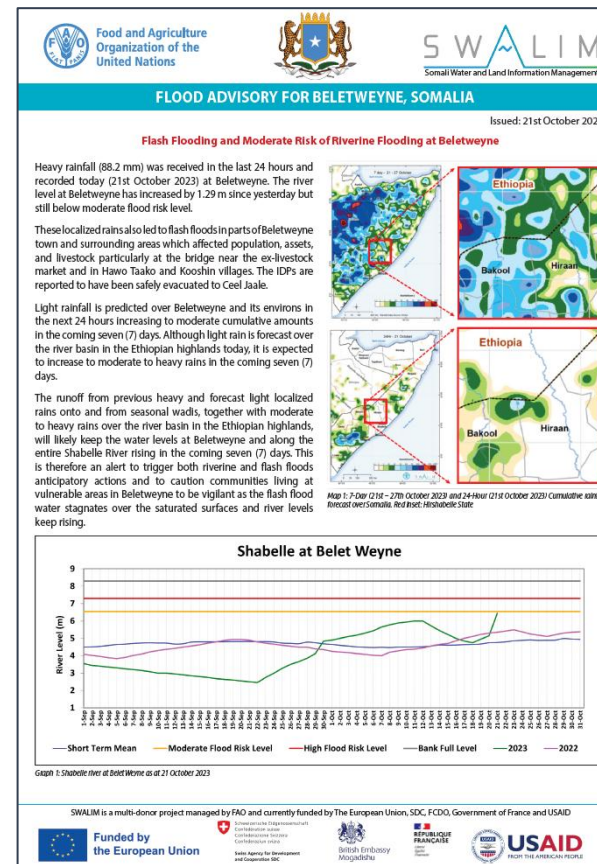
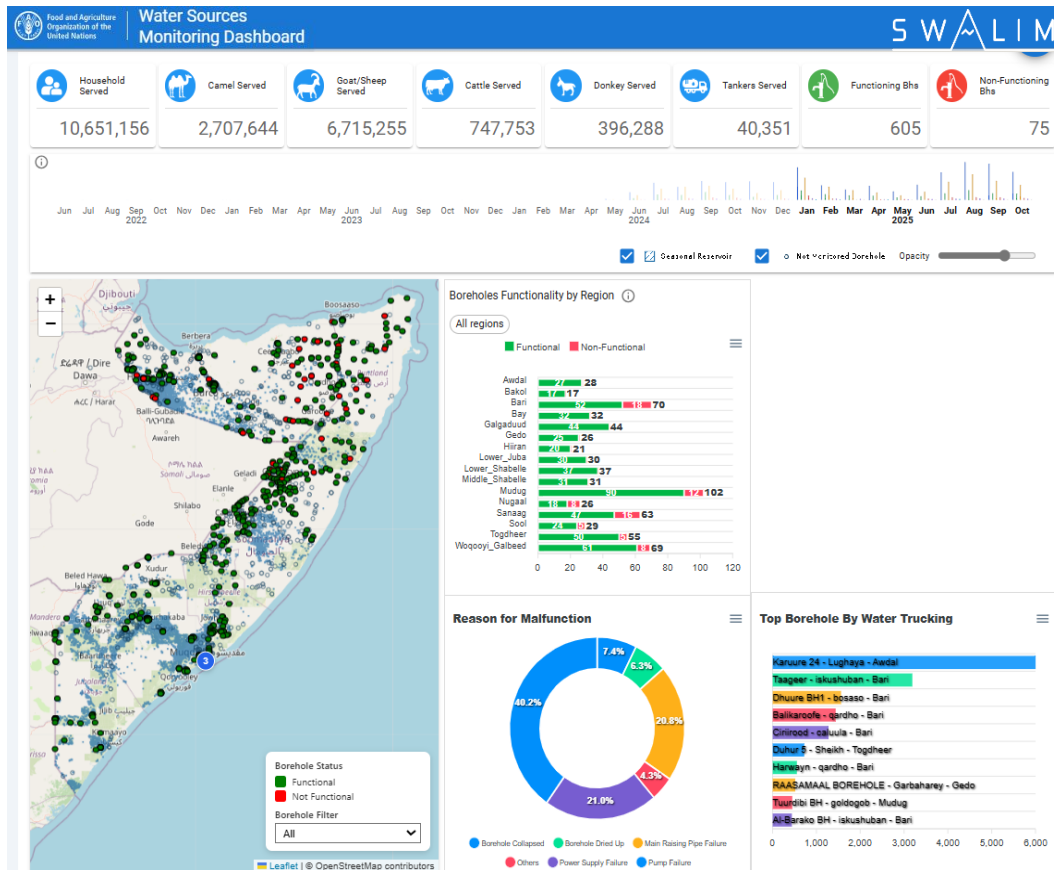
Land and water monitoring is the backbone of robust agriculture EW

FAO supports it by integrating satellite observations and field measurements for actionable information





What does it take to build EW systems that feed successful AA programs?



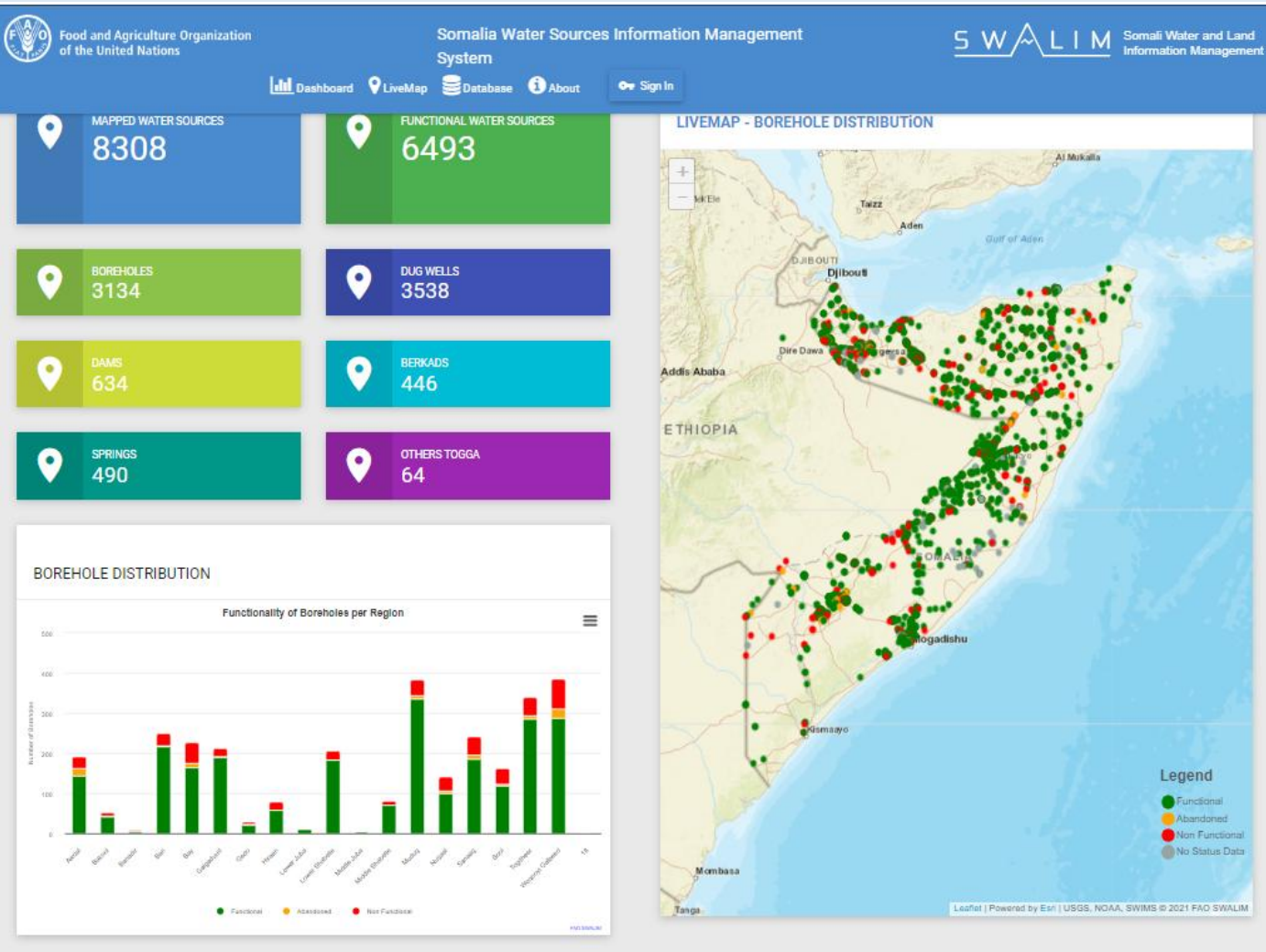
AWARENESS CAMPAIGNS

- ✓ Critical production decisions-climate smart agriculture
- ✓ Plant early maturing crops/ crop diversification etc.
- ✓ Engage in Flood recession Agriculture
- ✓ Stock up food reserves
- ✓ Enhanced grain storage
- ✓ Destocking
- ✓ Move to safer grounds



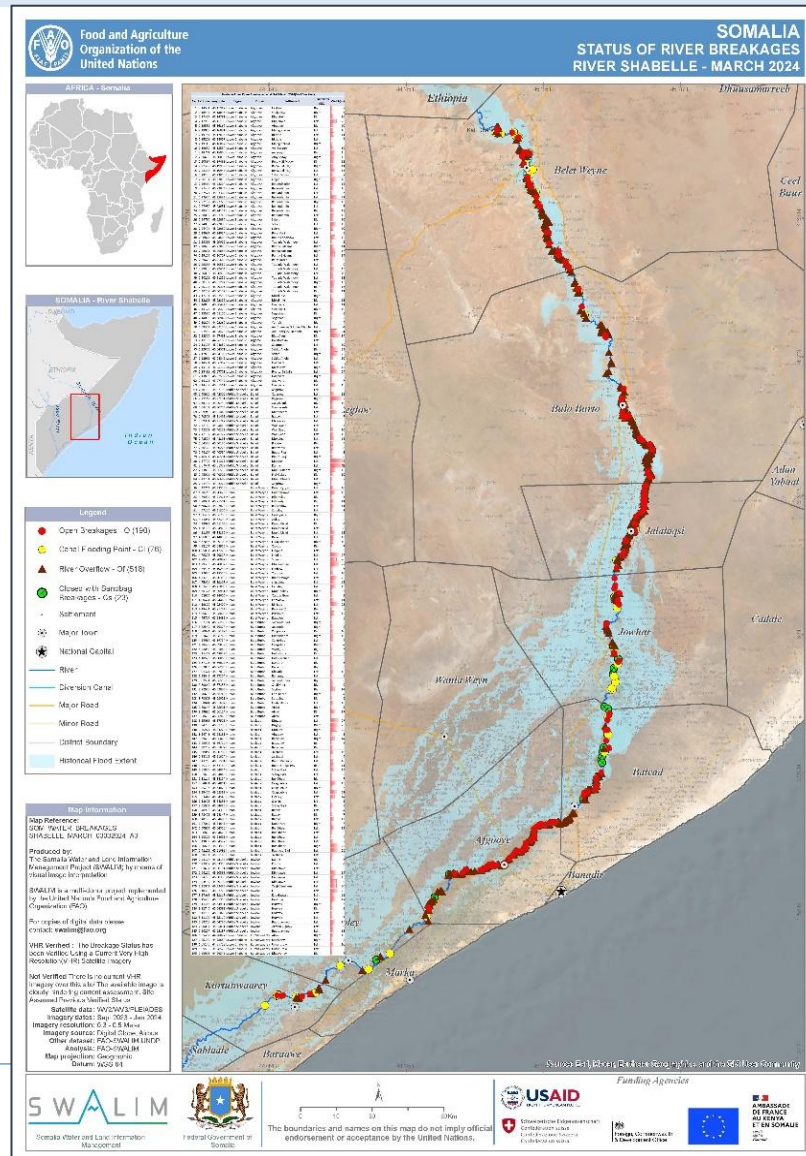


Strategic Water Sources Monitoring

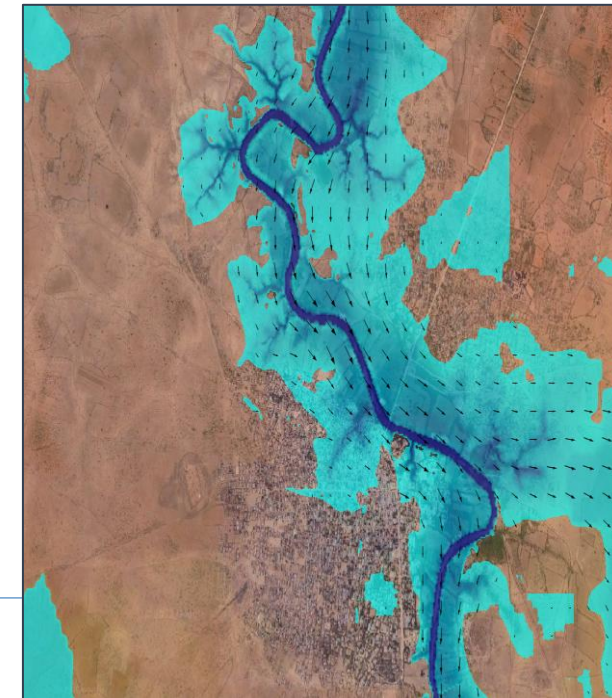
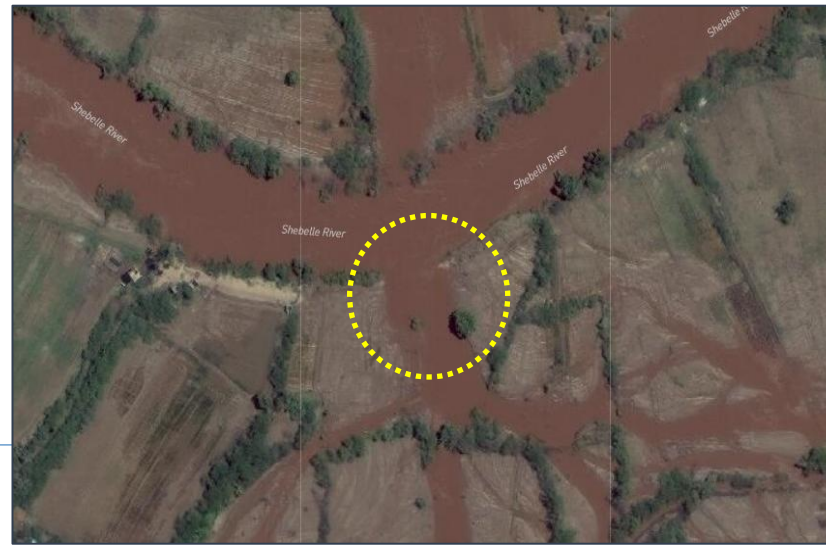
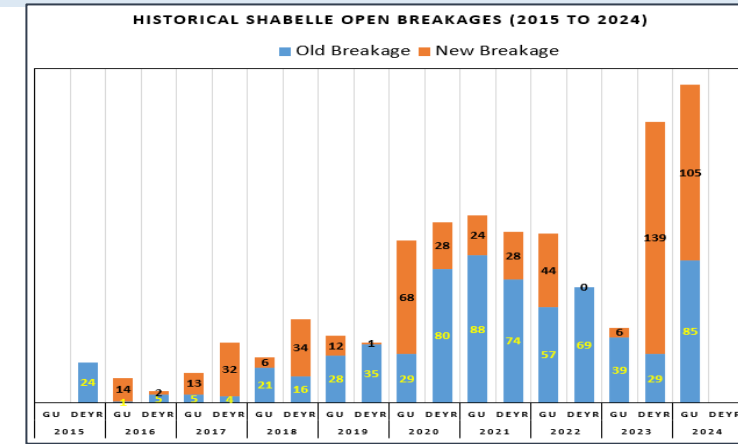


- ❖ National water sources database available in live map <https://swims.faoswalim.org/dashboard/view>
- ❖ Current database has over 8,300 strategic water sources: Boreholes, Shallow wells, springs, dams...
- ❖ Key parameters monitored: Functionality, yield, water quality (EC, pH), users, cost of water, etc.
- ❖ Detailed water quality tests on common compounds and heavy metals done for selected sources

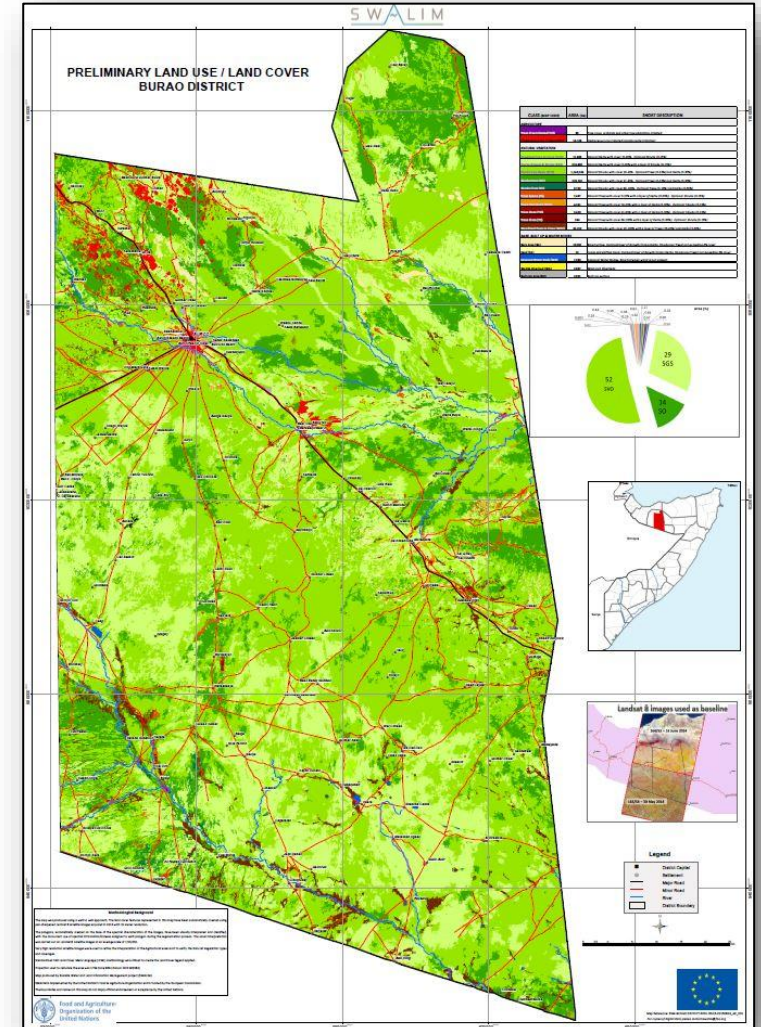
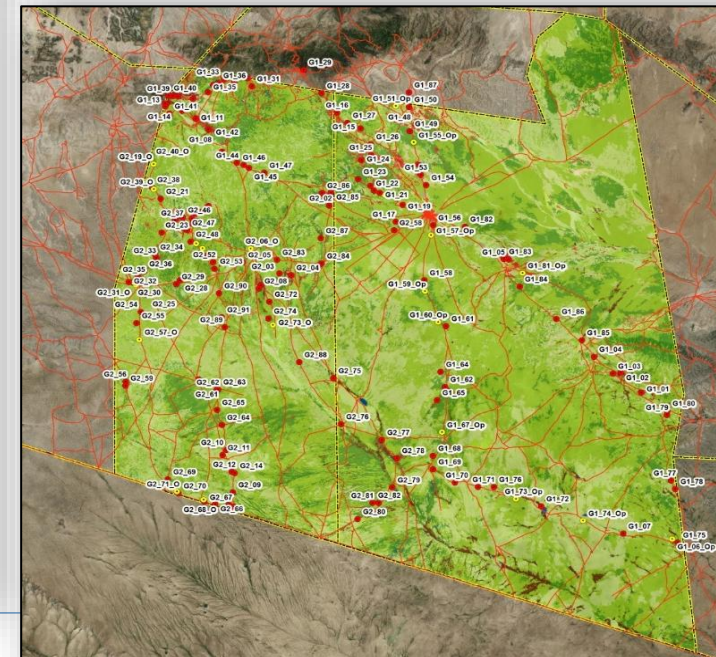
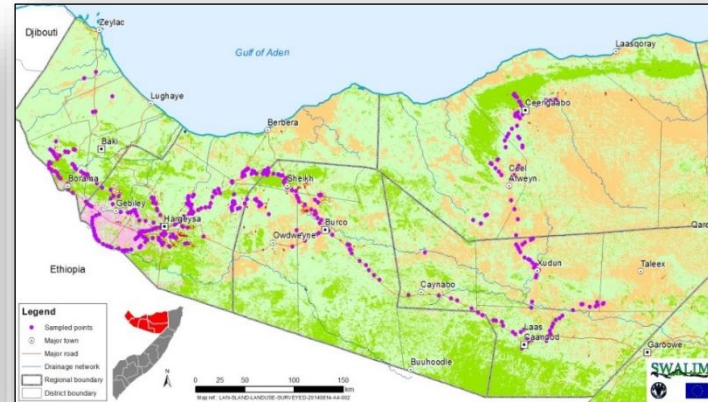
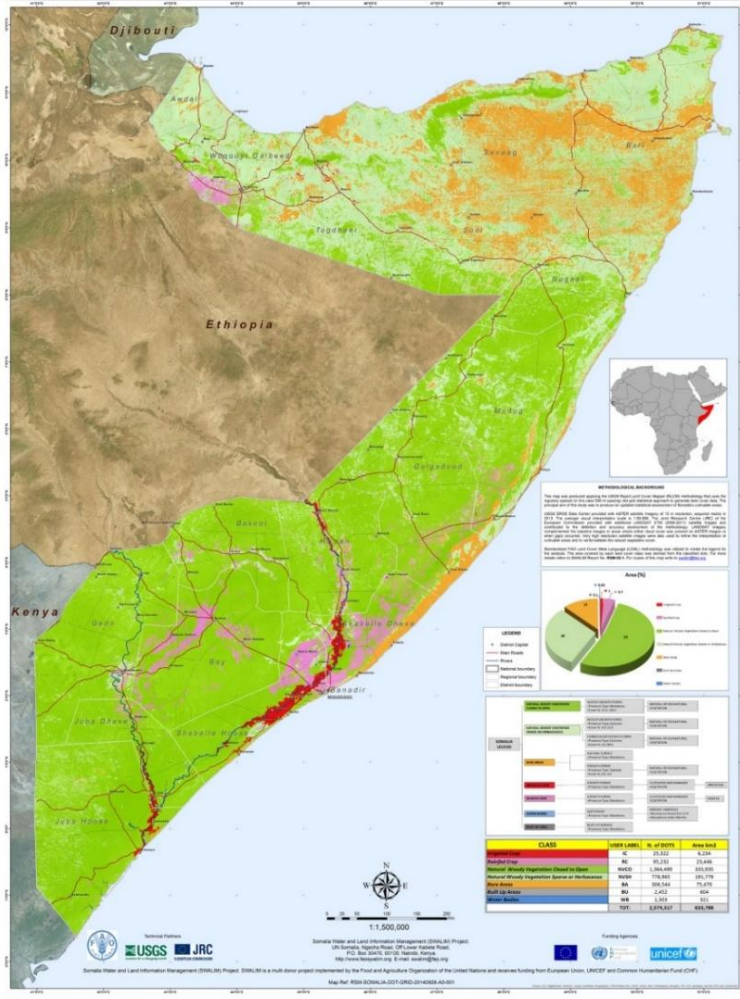
Flood preparedness and Early Action – Breakages Analysis



- Analysis of VHR satellite images performed twice a year to detect critical points along the river embankments
- Flood model along Shabelle and Juba
- Crucial datasets for flood preparedness and mitigation plans
- Implemented Digniin mobile platform for the timely dissemination of EW messages



Land Cover/Land Use mapping and field data survey



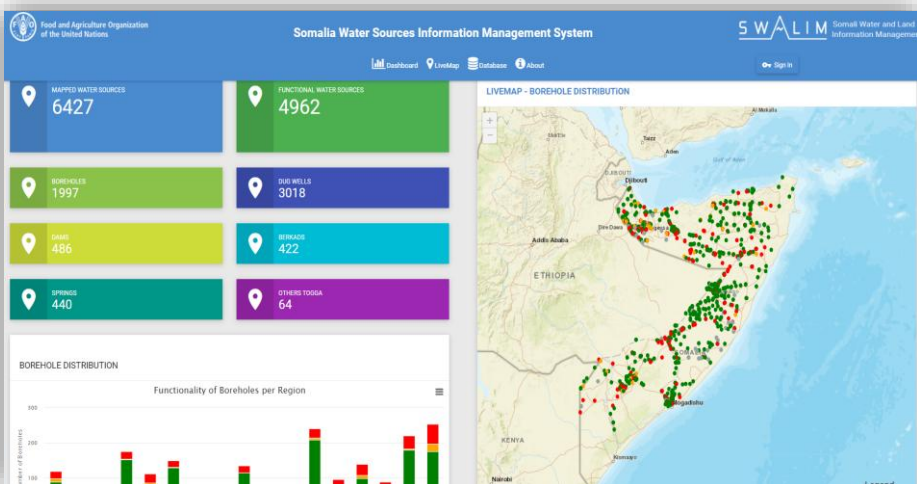
Land Cover map derived from a Dot-grid analysis

Land Cover maps derived from a wall-to-wall mapping

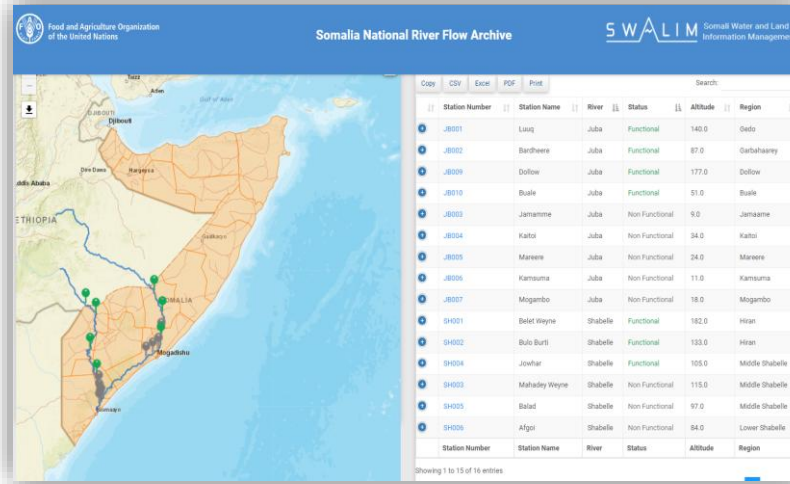


Information Dissemination: Online Systems

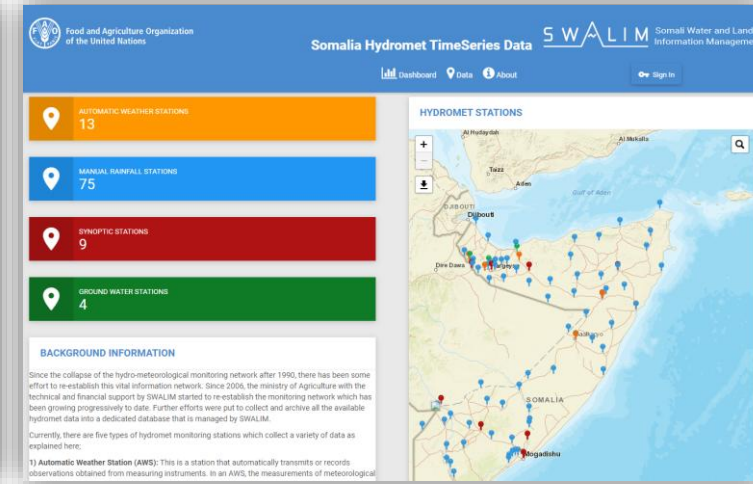
SOMALI WATER SOURCES LIVE MAP



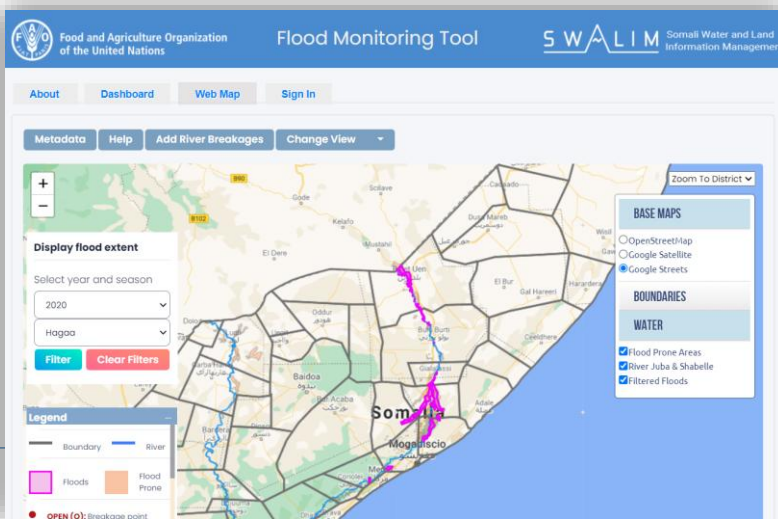
FLOOD & RIVER FLOW ARCHIVE



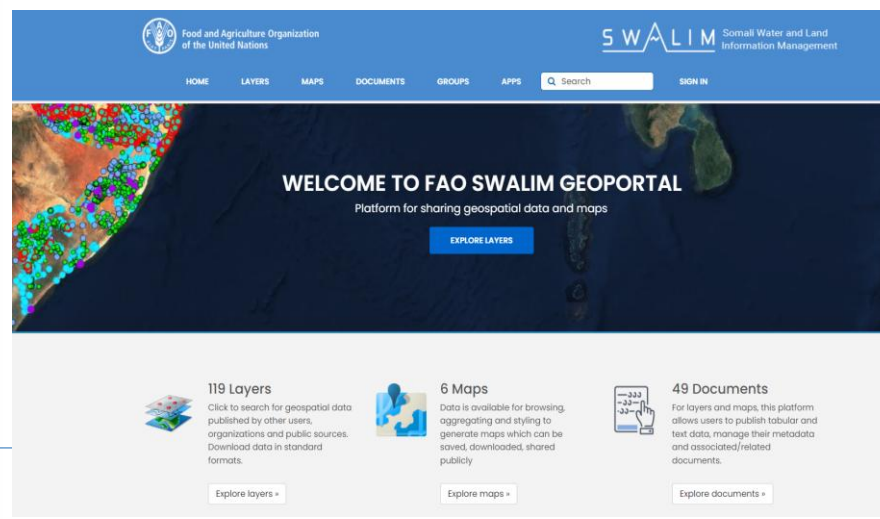
CLIMATE AND GROUND WATER SERIES DATA



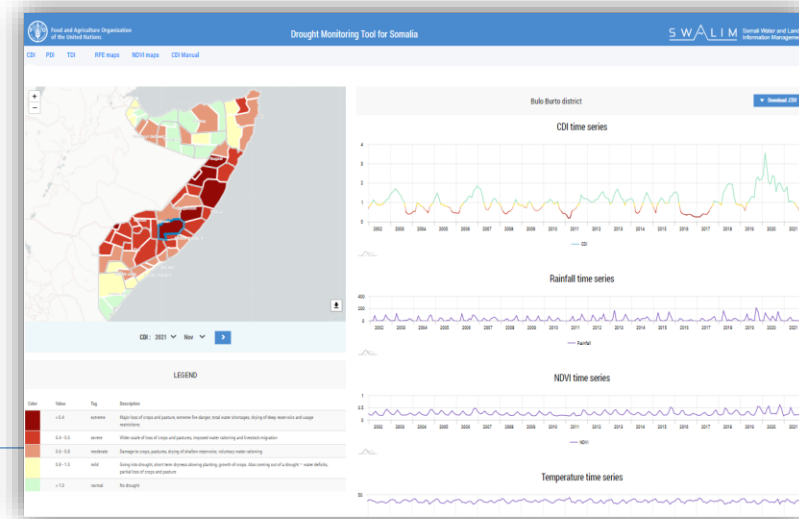
FLOOD MONITORING TOOL



GEOSPATIAL DATA PORTAL



SOMALI DROUGHT MONITORING TOOL





Training Program to transfer capacities to SWALIM line ministries



MAPATHONS

For effective early warning, early action in the Horn of Africa

15th Aug - 15th Oct 2023

Inviting **VOLUNTEERS** to map the anticipated impact of El Nino on vulnerable communities in flood prone areas of Kenya, Ethiopia and Somalia, to inform early action.

- Register using the QR code or this link: <https://forms.gle/4Vv5wNCy89Z2z67>
- Meeting link will be shared after registration
- Website: <https://www.faoswalim.org>
- Email: swalim@fao.org



 Food and Agriculture Organization of the United Nations

 Open Mapping Hub
EASTERN & SOUTHERN AFRICA



MAPATHON

For effective early warning, early action in Somalia

21 Nov to 04 Dec 2022

... We are inviting **VOLUNTEERS** to map vulnerable communities in Jowhar and Beletweyne towns at risk of flooding

Register using QR code or this link: <https://bit.ly/3Spvb5X>

Meeting link will be shared after registration

<https://www.faoswalim.org> : swalim@fao.org





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Thank you!

swalim@fao.org

